FIITJEE

JEE (Advanced)-2018

ALL INDIA TEST SERIES

PART TEST – III

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C.**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section–A (01 – 08, 21 – 28, 41 – 48)** contains 24 multiple choice questions which have one or more correct answer. Each question carries **+4 marks** for correct answer and **–2 marks** for wrong answer.

   Section–A (09 – 12, 29 – 32, 49 – 52) contains 12 paragraphs with each having 2 questions with one correct answer. Each question carries **+4 marks** for correct answer and **–2 marks** for wrong answer.

2. **Section–C (13 – 20, 33 – 40, 53 – 60)** contains 24 Numerical based questions with answers as numerical value from 0 to 9 and each question carries +4 marks for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

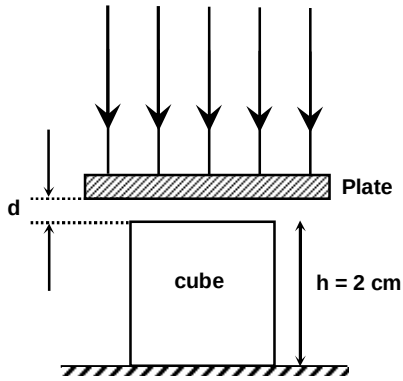
One OR More Than One Choice Type

This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

1. Choose the correct statement(s).
   (A) The image of sun (consider point source of light) can never be produced at center of glass sphere, (assuming the space between sphere and sun is filled with air).
   (B) If a lens having $\mu = 1.5$ has focal length 20 cm in air, then its focal length will be 60 cm when it is immersed in water of refractive index $\frac{4}{3}$
   (C) The laws of reflection and refraction are true for all surface and pairs of media at the point of incidence.
   (D) The thick lenses give coloured images due to dispersion of light.

2. When a yellow light of wavelength 6000 Å from a sodium lamp falls on a certain photocell, a negative potential of 0.30 volt is needed to stop all the electrons from reaching the collector. Then choose correct option(s).
   (A) Work function of the given photocell is 2.5 eV approximately.
   (B) Work function of the given photocell is 1.8 eV approximately.
   (C) A negative potential of 0.6 Volt will be needed to stop all the electrons if light of wavelength $\lambda = 4000$ Å is used
   (D) A negative potential of 1.3 Volt will be needed to stop all the electrons if light of wavelength $\lambda = 4000$ Å is used

3. An electromagnetic radiation of wavelength ranging between 400 nm and 1150 nm (for which the plate is penetrable) incident perpendicularly on the plate from above is reflected from both the air surfaces and interferes. In this range only two wavelength give maximum reinforcements, one of them is $\lambda = 400$ nm. (refractive index of air = 1 and the coefficient of linear thermal expansion of cube = 8×10^{-6} °C^{-1}). The distance of bottom of cube from plate does not change during warming up. Choose the correct option(s)



   (A) The change in temperature of cube so that it would touch the plate is 6.2°C
   (B) The change in temperature of cube so that it would touch the plate is 3.1°C
   (C) The second wavelength, which gives maximum reinforcement is $\lambda = \frac{2000}{3}$nm
   (D) The second wavelength, which gives maximum reinforcement is $\lambda = \frac{4000}{3}$nm

Space for Rough work

4. A certain ideal spring stretches 20 cm when m_1 = 40 gm mass is hung from it. If a mass of $m_2 = \frac{2205}{32}$gm gm is hung instead of 40 gm at its end pulled out 20 cm from equilibrium and released at t = 0 sec. Choose the correct option(s). (g = 10 m/s^2)

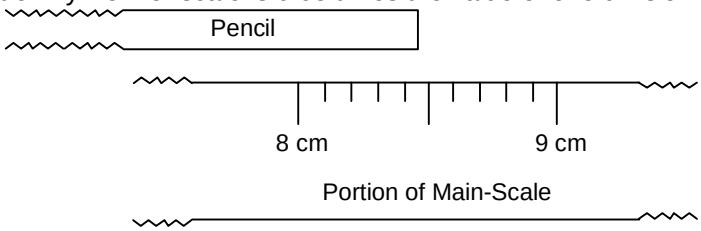
(A) The magnitude of acceleration of the m_2 mass when it is 2.5 cm from the equilibrium position is $\frac{64}{45}$ m/s^2

(B) The magnitude of acceleration of the m_2 mass when it is 2.5 cm from the equilibrium position is $\frac{32}{45}$ m/s^2

(C) At $t = \frac{\pi}{8}$ sec. m_2 mass is 10 cm from equilibrium position.

(D) At $t = \frac{\pi}{2}$ sec. m_2 mass is 10 cm from equilibrium position.

5. I am a Vernier Calliper. I have a task to measure the length of a pencil as shown in the figure. The least count of my Vernier scale is 0.05 times the value of one division of main-scale.



My Vernier Scale has n-divisions and coincides with (n – 4) divisions of main scale. While measuring the length of pencil, the $\frac{n}{5}$th division of Vernier-Scale exactly coincide the main-scale.

Now choose the correct option(s)

(A) The value of n is 40

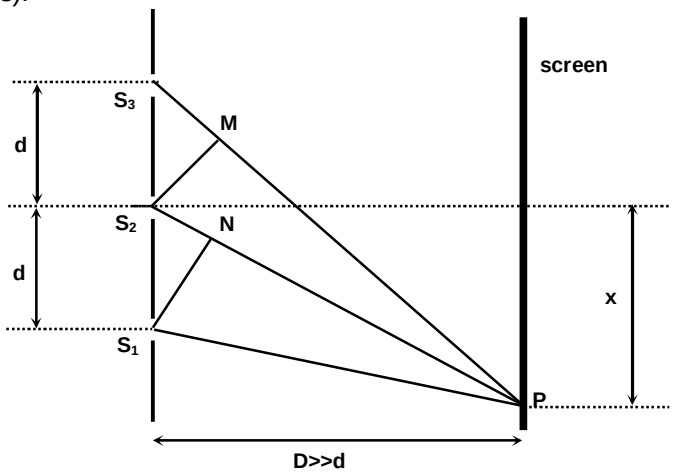
(B) The least count of Vernier scale is 0.005 cm

(C) The length of Pencil is 8.44 cm

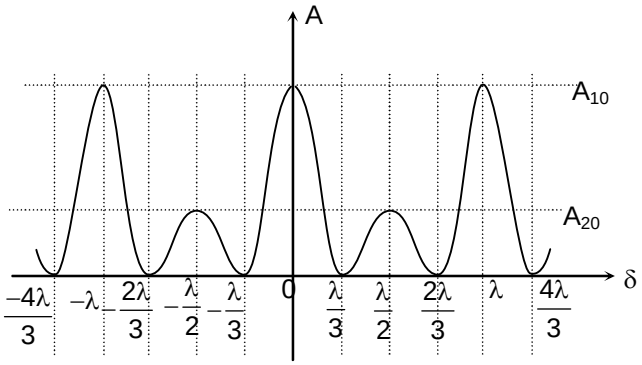
(D) The length of Pencil is 8.48 cm

Space for Rough work

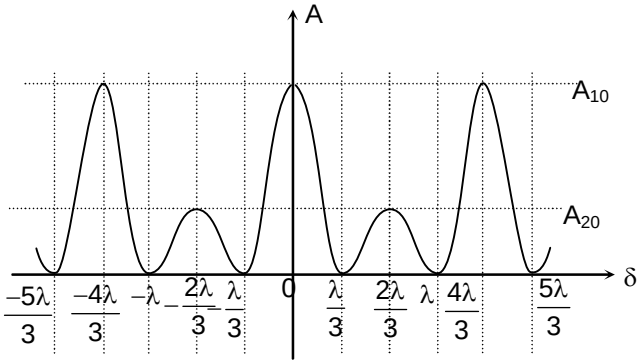
6. A triple slit experiment is performed as shown in the figure with monochromatic light of amplitude a_0. $S_3M \approx S_2N = \delta$. The graph between resultant amplitude (A) versus δ is plotted. Choose the correct option(s).



(A)



(B)



(C) $\frac{A_{10}+A_{20}}{A_{10}-A_{20}}=2$, where A_{10} and A_{20} are the values shown in option A.

(D) $\frac{A_{10}+A_{20}}{A_{10}-A_{20}}=5$, where A_{10} and A_{20} are the values shown in option A.

Space for Rough work

7. One end of thin aluminium rod (cross-sectional Area = 10^{-6} m^2) is bult welded to the end of copper rod with same diameter. Both rods are very long and under uniform tension of 900 N. Given that

$Y_{(Cu)} = 1.6 \times 10^{11}$ Nm^{-2} $Y_{(Al)} = 0.9 \times 10^{11}$ $N\,m^{-2}$

$\rho_{(Cu)} = 8.1 \times 10^3$ $kg\,m^{-3}$ $\rho_{(Al)} = 2.5 \times 10^3$ $Kg\,m^{-3}$

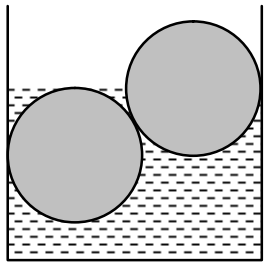
(A) The value of reflection factor $\frac{A_r}{A_i}$ for longitudinal pulse approaching the junction along copper rod is 0.15

(B) The value of reflection factor $\frac{A_r}{A_i}$ for longitudinal pulse approaching the junction along copper rod is approximately 0.3

(C) The value of reflection factor $\frac{A_r}{A_i}$ for transverse pulse approaching the junction along copper rod is 0.15

(D) The value of reflection factor $\frac{A_r}{A_i}$ for transverse pulse approaching the junction along copper rod is approximately 0.3

8. There are two cylinder-shaped wooden billets, each having a mass of mass 45 kg, in a vertical wall sewage, which contains water in it. The two billets have the same size and the same material; they touch each other and the walls of the sewage. One of them is totally under the water, whilst only half of the other one is immersed into the water. Friction is negligible everywhere. (density of water is 1000 kg/m^3 and $g = 10$ m/s^2). Choose the correct option(s)



(A) Density of the wood is 750 kg/m^3

(B) Density of the wood is 800 kg/m^3

(C) The forces exerted by the billets on the vertical walls is 260 N approximately

(D) The forces exerted by the billets on the vertical walls is 320 N approximately

Space for Rough work

Comprehension Type

This section contains **2 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

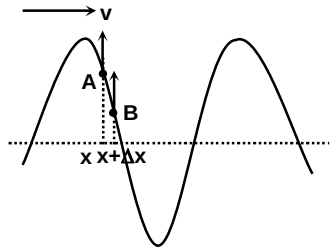
Paragraph for Question Nos. 09 and 10

Consider a wave sent down a string in the positive direction whose equation is given as

$$y = y_0 \sin\left[\omega\left(t - \frac{x}{v}\right)\right]$$

The wave is propagated along because each string segment pulls upward and downward on the segment adjacent to it at a slightly larger value of x and, as a result does work upon the string segment to which wave is travelling.

For example, the portion of string at point A is going upward, and will pull the portion at point B upward as well. In fact, at any point along the string, each segment of the string is pulling on the segment just adjacent and to its right, causing the wave to propagate. It is by this process that the energy is sent along the string.

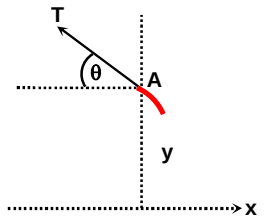


Now we try to calculate how much energy is propagated down the string per second

$$T_y = T\sin\theta \approx T\tan\theta \Rightarrow T_y = -T\frac{\partial y}{\partial x}$$

(The negative sign appears because as shown in the figure-II, the slope is negative) the force will act through a distance

$$dy = v_y dt = \frac{\partial y}{\partial t}dt$$



Therefore work done by force in time dt is

$$dW = T_y dy = -T\left(\frac{\partial y}{\partial x}\right)\left(\frac{\partial y}{\partial t}\right)dt \Rightarrow dW = \frac{\omega^2 y_0^2 T}{v}\cos^2\left[\omega\left(t - \frac{x}{v}\right)\right]dt \quad \ldots(A)$$

Now answer the following questions.

9. The average power transmitted down the string (i.e. the average energy transfer or sent down it per second) is

(A) $\frac{2\omega^2 y_0^2 T}{v}$ (B) $\frac{\omega^2 y_0^2 T}{v}$

(C) $\frac{\omega^2 y_0^2 T}{2v}$ (D) $\frac{\omega^2 y_0^2 T}{4v}$

10. How much average power is transmitted dawn a string having density $\mu = 5 \times 10^{-4}$ kg/m and T = 5 N by a 200 Hz vibration of amplitude of 0.20 cm.

(A) 0.08 Watt (B) 0.16 Watt

(C) 0.32 Watt (D) 0.64 Watt

Space for Rough work

Paragraph for Question Nos. 11 and 12

ABC is an isosceles right angled triangular reflecting prism of average refractive index μ . When incident rays on face AB are parallel to face BC , then they emerges from Face AC , which are also parallel to Face BC as shown in figure-I. The prism capable to do so , known as **Dove Prism.** In figure-II, the **Dove Prism** is used for dispersion of incident light containing red colour and violet colour only. The red colour and violet colour lights are separated(displaced) on screen by a distance $\Delta\ell$. In reality , each ray of any colour has some width, which can be designated as d. It is clear that an observer can distinguish the red and violet rays that emerges from prism only if $\Delta\ell \geq d$ .Otherwise the bundles of rays will overlap and mix.

[Given for **Dove Prism** in figure-II: $\mu_R = \sqrt{\frac{5}{2}}$, $\Delta\mu = \mu_V - \mu_R = 0.02$, $\sqrt{5} = 2.25$, EF = 1m and AB =4cm]

Now answer the following questions.

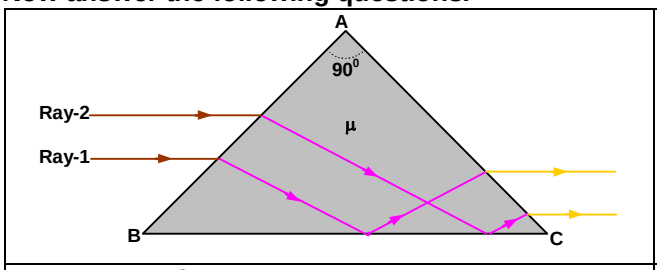


Figure-I

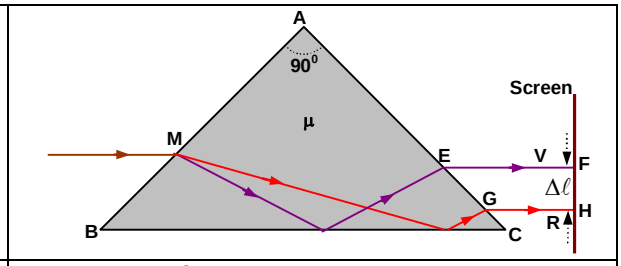


Figure-II

11. Choose the **INCORRECT** option.
(A) As per figure-I, the average refractive index of **Dove Prism** may be greater than 1.
(B) As per figure-I, the average refractive index of **Dove Prism** may be greater than $\sqrt{2}$.
(C) As per figure-II, the displacement $\Delta\ell$ depends upon average refractive index μ and length of AB.
(D) As per figure-II, the displacement $\Delta\ell$ depends upon average refractive index μ and length of EF.

12. Find the maximum value of width d of bundle of rays that can be resolved at the outlet of **Dove Prism** as shown in the figure-II.
(A) $2.25\times10^{-3}\,m$ (B) $2.25\times10^{-4}\,m$
(C) $1.125\times10^{-3}\,m$ (D) $1.125\times10^{-4}\,m$

Space for Rough work

SECTION – C

(One Integer Value Correct Type)

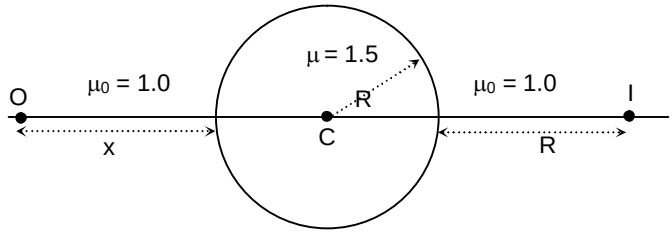
This section contains **8 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive)

13. If the dimension of a physical quantity $\frac{EB}{\mu}$ is $M^aL^bT^c$. Find the value of $\left(\frac{b-c}{a}\right)$. Here

$E \Rightarrow$ magnitude of electric field
$B \Rightarrow$ Magnitude of magnetic field
$\mu \Rightarrow$ Permeability of medium

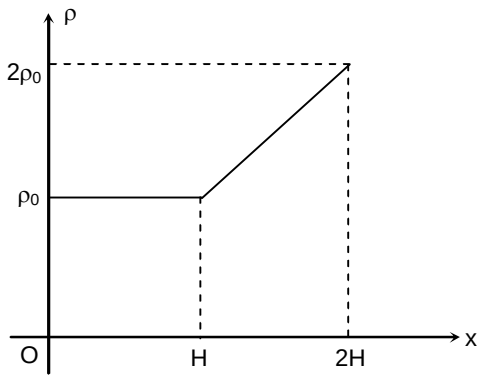
14. A 200 cm length of wire weighs 0.6 gm. If the tension in the wire caused by a 500 gm mass hanged from it. If the approximate wavelength of a wave of frequency 400 Hz sent down by it is $2^n \times 10^{-2}$ m. Find the value fo n. ($g = 9.8$ m/s^2)

15. A glass sphere of refractive index 1.5 forms the real image of object O at point I as shown in the figure when kept in air of refractive index 1. If the value of x is kR. Find k.

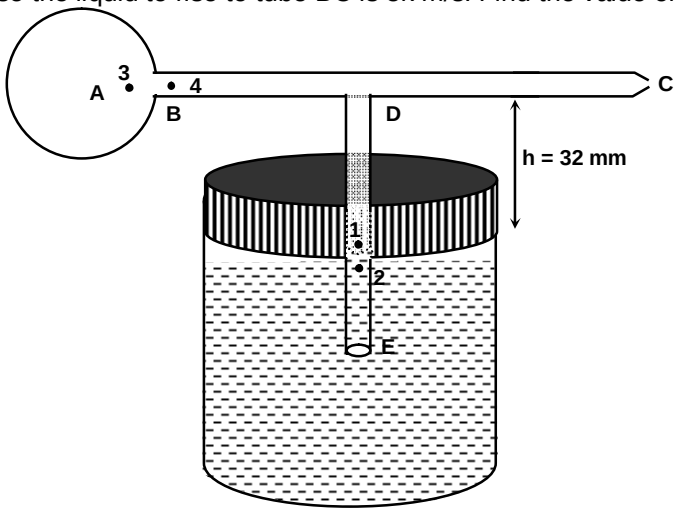


Space for Rough work

16. A large cylindrical container with open top contains a liquid upto height (2H = 8m) and whose density varies as shown in the figure. Atmospheric pressure is 1 atm (= 10^5 N/m^2)
Find the pressure at bottom of cylindrical container in atm.
(given $\rho_0 = 10^3 \text{ Kg/m}^3$, and $g = 10 \text{ m/s}^2$).



17. The figure shows a crude type of perfume atomizer. When bulb at A is compressed, air flows swiftly through the tiny BC with uniform speed v, there by causing a reduced pressure at the position of vertical tube DE. The liquid of density 500 kg/m^3, then rises in the tube, enters tube BC and sprayed out. When bulb is in natural position the air in the bulb and tube are at atmospheric pressure $P_{0 = 10^5}$ N/m^2. When bulb A is compressed, it creates an excess pressure $\Delta p = 0.001\ P_0$ inside the bulb A. Density of air is 1.3 kg/m^3. If the magnitude of minimum value of speed v, required to cause the liquid to rise to tube BC is 5k m/s. Find the value of k. (g = 10 m/s^2).



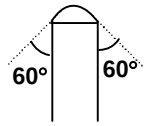
Space for Rough work

18. A hydrogen atom in ground state, moving with speed v collides with another hydrogen atom in ground state at rest. If $v \le v_0 = \sqrt{\frac{kE_0}{m_H}} \approx a \bullet bc \times 10^n \, m/s,$ then the collision is elastic. Here a, b and c are whole number, less then 9. Find the value of $\frac{a \times b \times c}{k \times n}$.

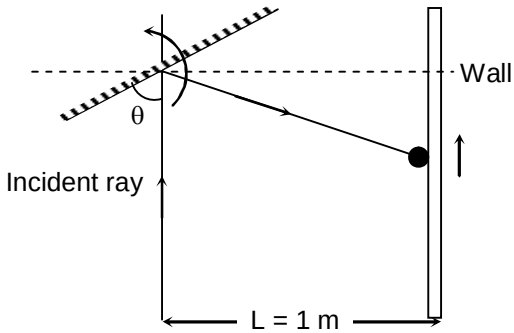
$\frac{me^4}{2\hbar^2} = E_0 = 13.6\,eV/atom = 2.18 \times 10^{-18}$ J/atom $\Rightarrow$ ionisation energy of H-atom

$\Rightarrow \quad m_H = 1.67 \times 10^{-27} kg \Rightarrow$ Mass of hydrogen atom.

19. A soap bubble is being blown on a tube of radius 1 cm. The surface tension of the soap solution is 0.05 N/m and the bubble makes an angle of 60° with the tube in equilibrium state as shown. If the excess of pressure over the atmospheric pressure in the tube in Pascal (Pa) is x, find x



20. A light ray is incident on a rotating mirror, which is rotating with angular speed 3 rad/s as shown in figure. The speed of bright spot on the wall when $\theta = 30°$.



Space for Rough work

Chemistry

PART – II

SECTION – A

One OR More Than One Choice Type

This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

21. Which of the following statement is/are correct?

(A) $CrI_3 + KOH + Cl_2 \longrightarrow K_2CrO_4 + KCl + KIO_4 + H_2O$

In the balanced chemical reaction, stoichiometric coefficient of KOH is 64.

(B) $(NH_4)_2 Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$

n-Factor of $(NH_4)_2Cr_2O_7$ is 12.

(C) $\left[Fe(CN)_6\right]^{3-} \longrightarrow Fe^{3+} + CO_2 + NO_3^-$

n-Factor for $[Fe(CN)_6]^{3-}$ is 60.

(D) $MnO_4^{2-} \xrightarrow{H^+} MnO_4^- + MnO_2$

Eq. wt. of MnO_4^{2-} is 178.5.

22. Which of the following on reaction with H_2S will produce metal sulphide?

(A) $COCl_2$ (B) $CuCl_2$

(C) $\left[Cd(CN)_4\right]^{2-}$ (D) $\left[Zn(OH)_4\right]^{2-}$

23. A dilute solution contains m mol of solute A in 1 kg of a solvent with molal elevation constant K_b. The solute dimerises in solution as

$2A \rightleftharpoons A_2$

Select the correct expression of equilibrium constant 'K' for the dimer formation. (assume molarity = molality)

(A) $K = \dfrac{K_b(K_b m - \Delta T_b)}{(K_b m + 2\Delta T_b)^2}$ (B) $K = \dfrac{\alpha}{2m(1-\alpha)^2}$

(C) $K = \dfrac{K_b(K_b m - \Delta T_b)}{(2\Delta T_b - K_b m)^2}$ (D) $K = \dfrac{2(K_b m - \Delta T_b)}{K_b m}$

24. Select the correct statement for cloud seeding:

(A) It promotes rainfall.

(B) It suppress fog.

(C) Most commonly used chemical for cloud seeding is silver iodide because it has a crystalline structure to that of ice.

(D) It helps in snow fall.

Space for Rough work

25. Which of the following combination, of compounds/elements and its packing fraction is true? Assume all cations or anions are ideally fitted in the voids without any distortion in lattice.

(A) Na_2O, $\phi = \frac{\pi}{3\sqrt{2}}\left\{1+2\times(0.225)^3\right\}$ (B) Cr_2O_3, $\phi = \frac{\pi}{3\sqrt{2}}\left\{1+\frac{2}{3}\times(0.414)^3\right\}$

(C) Fe_2O_3, $\phi = \frac{\pi}{3\sqrt{2}}\left\{1+(0.225)^3\right\}$ (D) Diamond, $\phi = \frac{\pi}{3\sqrt{2}}\left\{1+(0.414)^3\right\}$

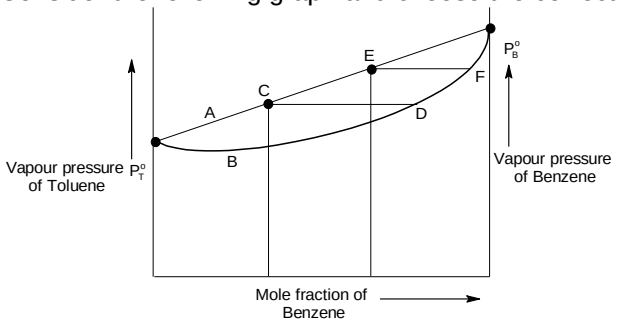
26. In which of the following ores at least one metal is in +3 oxidation state and crystal system is orthorhombic.

(A) Haematite (B) Copper pyrite
(C) Diaspore (D) Aragonite

27. Which of the following IUPAC names are correct?

(A) $K_2[OsCl_5N]$: Potassium pentachloridonitridoosmate(VI)
(B) $[Fe(NH_4C_5 - C_5H_4N)_3]Cl_2$: Tris(bipyridine)iron(II) chloride
(C) $[Fe(C_5H_5)_2]$: Bis(η^5-cyclopentadienyl)iron(II)
(D) $[Pt(py)_4][PtCl_4]$: Tetrapyridineplatinum(II) tetrachloridoplatinate(II)

28. Consider the following graph and choose the correct option



(A) A represent vapour composition and B represent liquid composition.
(B) At the state E of the system the vapour phase just start forming and its vapour phase composition will correspond to the point C.
(C) At the state C of the system the vapour phase just start forming and its vapour phase composition will correspond to the point D.
(D) A and B both represent vapour composition.

Space for Rough work

Comprehension Type

This section contains **2 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Question Nos. 29 to 30

A metal crystallizes in a closed packed structure, the atoms stack together occupying maximum space and leaving minimum vacant space. Each sphere in the first layer is in contact with six neighbours. Now while arranging the second layer on the first layer, spheres are placed in depression of first layer. Spheres of third layer lie in the depression of second layer.
This repeating arrangement of atoms produces a giant lattice.

29. If one of the edge length of unit cell is 'a' what is the shortest distance between two atoms?

(A) $\frac{\sqrt{3}a}{4}$ (B) $\frac{a}{4}$ or $\frac{a}{3}$

(C) a or $\frac{a}{\sqrt{2}}$ (D) $\frac{a}{2}$

30. If spheres of third layer do not lie directly over the atoms of first layer, then density of unit cell is:

(Given: atomic weight of metal is 197u and a = 4.07 $\overset{o}{A}$)

(A) 86.5 g/cc (B) 19.4 g/cc
(C) 56.2 g/cc (D) 40.5 g/cc

Space for Rough work

Paragraph for Question Nos. 31 to 32

Stereoisomerism occurs when the bond connectivity is same but the spatial arrangement of more than one type is possible. Stereoisomerism finds vast use in organic chemistry, mainly biochemistry as all the biomolecules, are generally stereospecific in their actions. Stereoisomerism is limited not only to organic compounds but also to inorganic compounds, mainly in co-ordination complexes. Stereoisomers include *cis* and *trans* isomers, chiral isomers, compound with different conformation of chelate rings and other isomers that differ only in the geometry of attachment to the metal ion.

31. Select the incorrect statement:
(A) trans form of $[CoCl_2(trien)]^+$ has 3 coplanar chelate ring.
(B) $[Pt(py)(NH_3)(NO_2)(Cl)(Br)(I)]$ has 15 geometrical isomers.
(C) Carboxylic group of salicylate ion is *cis* and *trans* to tertiary nitrogen in $[Co(tren)(sal)]^+$.
(D) $[Pt(NH_3)_2(py)_2Cl_2]^{2+}$ complex has 5 non chiral isomer.

32. Select the correct option for tetraamminedichlorocobalt(III) ion, the *trans* isomer
(A) does not have any symmetry
(B) has plane of symmetry
(C) *trans* form is meso form
(D) *trans* form exists as a pair of enantiomer

Space for Rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **8 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive)

33. The enthalpy change involved in oxidation of glucose is - 2880 kJ mol^{-1}, 25% of this energy is available for muscular work. if 100 kJ of muscular work is needed to walk one km, the maximum distance (km) that a person will be able to walk after taking 150 g of glucose is……

34. 1 litre solution containing 4.5 millimoles of $Cr_2O_7^{--}$ and 15 millimoles of Cr^{3+} shows a pH of 2.0. The potential of the reduction half reaction is approximately x V. Then [x] is ……., where [ ] is GIF. $\left(E^o \text{ of } Cr_2O_7^{2-}/Cr^{3+} \text{ is } 1.33 \text{ V}\right)$

35. Assuming 100% polymerization of an organic compound in its aqueous solution, the number of moles of organic compound undergoing polymerization containing 9.4 g of organic compound per 100 g of the solvent is…..... Freezing point depression is 0.93 K, K_f of water is 1.86 K kg mol^{-1} and molecular weight of organic compound is 94u.

36. A gas consists of 5 molecules with a velocity of 5 ms^{-1}, 10 molecules with a velocity of 3 ms^{-1} and 6 molecules with a velocity of 6 ms^{-1}. The ratio of average velocity and rms velocity of the molecules is 9.5×10^{-x}. Where x is …..

Space for Rough work

37. Number of S – S linkage in Marshall acid = a
Number of P – P linkage in Pyrophosphoric acid = b
Number of O – O linkage in Caro acid = c
Number of C = C linkage in Carbon suboxide = d
Find out the value of a + b + c + d?

38. In an ionic solid $r_c = 1.6\,\mathring{A}$ and $r_a = 1.864\,\mathring{A}$. Find edge length of cubic unit cell in $\mathring{A}$

39. Percentage of silver in german silver…..

40. $A \xrightarrow{\text{KCN added slowly}} \underset{(\text{Reddish brown precipitate})}{B} \xrightarrow{\text{In excess KCN precipitate dissolves}} \underset{(\text{Yellow solution})}{C}$

$A \xrightarrow{K_4Fe(CN)_6} \underset{(\text{Intense blue precipitate})}{D} \xrightarrow{NaOH} (\text{Red precipitate})$

$A \xrightarrow{Na_2HPO_4} \underset{(\text{Yellowish white precipitate})}{E}$

How many products are correctly match?

$A = FeCl_3$	$A = FeCl_2$	$A = CuCl_2$	$B = Fe(CN)_3$
$B = K_3Fe(CN)_6$	$C = K_3Fe(CN)_6$	$C = K_4[Fe(CN)_6]$	$D = Fe_4[Fe(CN)_6]_3$
$D = Fe[Fe(CN)_6]$	$E = FePO_4$		

Space for Rough work

Mathematics

PART – III

SECTION – A

One OR More Than One Choice Type

This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

41. A is a matrix of order 3×3 and a_{ij} is its elements of ith row and j^{th} column. If $a_{ij} + a_{jk} + a_{ki} = 0$ holds for all $1 \le i, j, k \le 3$ then
(A) A is a non-singular matrix
(B) A is a singular matrix
(C) $\sum_{1\le i,\, j\le 3} a_{ij}$ is equal to zero
(D) A is a symmetric matrix

42. If x and y satisfy the equation $xy - 2x^2 - 9x + 3y - 16 = 0$, then
(A) number of ordered pair (x, y) is 4 where $x, y \in Z$
(B) number of ordered pair (x, y) is 1 where $x, y \in N$
(C) if $x \ge y$, $x, y \in N$ then number of ordered pair is zero
(D) if $x \le y$, $x, y \in N$ then number of ordered pairs are two

43. Let p, q be integers and let α, β be the roots of the equation $x^2 - 2x + 3 = 0$ where $\alpha \ne \beta$, if n = 0, 1, 2, 3, ... Let $a_n = p\alpha^n + q\beta^n$ where $n \in \{0, 1, 2, ...\}$ then the value of a_9 is equal to
(A) $3a_8 - 5a_7 + 3a_6$
(B) $a_7 - 6a_6$
(C) $a_7 - 12a_5 + 18a_4$
(D) $2a_8 + 3a_6$

44. For any two positive integers x and y $f(x, y) = \frac{1}{(x+1)!} + \frac{1}{(x+2)!} + \frac{1}{(x+3)!} + ... + \frac{1}{(x+y)!}$, then which of the following options is/are correct
(A) $f(x, y) \le \frac{1}{x}\left(\frac{1}{x!} - \frac{1}{(x+y)!}\right)$
(B) $\lim_{y\to\infty}\left(f(x, y)\right) \le \frac{1}{x!}$
(C) $f(x, x) < \frac{1}{(x-1)!}$
(D) f(2, 2) is equal to $\frac{2}{3}$

Space for rough work

45. Let a > 0, b > 0, c > 0 and a + b + c = 6 then $\frac{(ab+1)^2}{b^2}+\frac{(bc+1)^2}{c^2}+\frac{(ca+1)^2}{a^2}$ may be

(A) $\frac{75}{4}$ (B) 35

(C) 15 (D) 10

46. A and B play a game in which they alternately call out positive integers less than or equal to n, according to the following rules. A goes first and always calls out an odd number, B always calls out an even number and each player must call out a number which is greater than the previous number (except for A′s first turn). The game ends when one player cannot call out a number, then which of the following is/are correct?

(A) For n = 6, number of possible games is 8 (B) For n = 6, number of possible games is 10

(C) For n = 10, number of possible games is 55 (D) For n = 8, number of possible games is 21

47. Which of the following is/are correct?

(A) The point of intersection of $\overrightarrow{AB}$ and $\overrightarrow{CD}$ where A(4, 7, 8), B(–1, –2, 1), C(2, 3, 4) and D(1, 2, 5) is $\left(\frac{3}{2},\frac{5}{2},\frac{9}{2}\right)$

(B) If (α, β, γ) be a point on the plane 2x + 6y + 15z = 7 then the least value of $\alpha^2 + \beta^2 + 25\gamma^2$ is 7

(C) The circumcentre of the triangle formed by the points (3, 2, –5), (–3, 8, –5) and (–3, 2, 1) is the point (–1, 4, –3)

(D) The length of intercept on the line $\frac{x+1}{-1}=\frac{y-12}{5}=\frac{z-7}{2}$ by the surface $11x^2 – 5y^2 + z^2 = 0$ is $\sqrt{30}$ units

48. Consider all 10 digit numbers formed by using all the digits 0, 1, 2, 3, ..., 9 without repetition such that they are divisible by 11111, then

(A) the digit in tens place for smallest number is 6

(B) the digit in tens place for largest number is 3

(C) total numbers of such numbers is 3456

(D) total numbers of such numbers is 4365

Space for rough work

Comprehension Type

This section contains **2 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Question Nos. 49 to 50

Read the following write up carefully and answer the following questions:
Let $A(z_1)$, $B(z_2)$, $C(z_3)$ and $D(z_4)$ be the vertices of a trapezium in an Argand plane such that AB || CD. Let $|z_1 - z_2| = 4$, $|z_3 - z_4| = 10$ and the diagonals AC and BD intersect at P. It is given that $\text{Arg}\left(\frac{z_4 - z_2}{z_3 - z_1}\right) = \frac{\pi}{2}$ and $\text{Arg}\left(\frac{z_3 - z_2}{z_4 - z_1}\right) = \frac{\pi}{4}$

49. Which of the following option(s) is/are correct?

(A) Area of the trapezium ABCD is equal to $\frac{140}{3}$ sq. units

(B) Area of the trapezium ABCD is equal to $\frac{70}{3}$ sq. units

(C) Area of the triangle BCP is equal to $\frac{100}{21}$ sq. units

(D) Area of the triangle BCP is equal to $\frac{200}{42}$ sq. units

50. Which of the following option(s) is/are incorrect?

(A) value of $|CP - DP|$ is equal to $\frac{10}{\sqrt{21}}$ (B) PB : PD = 2 : 5

(C) PC : AC = 5 : 7 (D) PC : AC = 2 : 3

Space for rough work

Paragraph for Question Nos. 51 to 52

Read the following write up carefully and answer the following questions:
In the equation A + B + C + D + E = FG, where FG is the two digit number whose value is 10F + G and letters A, B, C, D, E, F and G each represents different digits and FG is as large as possible

51. A five digit number is made using digits A, B, C, D, E, F, G (repetition not allowed), then which of the following is incorrect

(A) probability that number made is divisible by 5 is $\frac{1}{7}$

(B) probability that number made is divisible by 3 is $\frac{2}{7}$

(C) probability that number made is divisible by 4 is $\frac{2}{7}$

(D) probability that number made is divisible by 4 is $\frac{1}{7}$

52. If a five digit number is made using the digits A, B, C, D, E without repetition then

(A) the probability that the number is divisible by 11 is $\frac{2}{5}$

(B) the probability that the number is divisible by 11 is $\frac{1}{5}$

(C) the probability that the number is divisible by 4 is $\frac{1}{4}$

(D) the probability that the number is divisible by 4 is $\frac{2}{5}$

SECTION – C

(One Integer Value Correct Type)

This section contains **8 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive)

53. If a, b, c are three positive real numbers then the minimum value of $\frac{a+3c}{a+2b+c}+\frac{4b}{a+b+2c}-\frac{8c}{a+b+3c}$ is $\alpha+\beta\sqrt{2}$ (where a, b $\in$ Z), then $|\alpha + \beta|$ is equal to _____

Space for rough work

54. If the quadratic equation $a_1x^2 - a_2x + a_3 = 0$ where $a_1, a_2, a_3 \in N$ has two distinct real roots belonging to the interval (1, 2), then least value of a_1 is _____

55. Let a_1, a_2, a_3, a_4, a_5 be five terms of an increasing geometric progression such that $a_1, a_2, a_3, a_4, a_5 \in N$, and $a_5 < 100$. Then the number of possible geometric progression (is/are) _____

56. If the coefficient of $x^{\frac{n^2+n-18}{2}}$ in $(x-1)(x^2-2)(x^3-3)(x^4-4)\ldots.(x^n-n)$, where $(n \geq 8)$ is k, then k is equal to _____

57. If a, b, c are three positive numbers then the minimum value of $\frac{a^4+b^6+c^8}{\left(ab^3c^2\right)2\sqrt{2}}$ is equal to _____

58. Let α^k when k = 0, 1, 2, 3, 4, ..., 253 are 254th roots of unity then the unit digit of $\sum_{k=0}^{253}\left|z+\alpha^k z^2\right|^2$ is _____ (where $z = e^{i\left(\frac{2\pi}{7}\right)}$)

59. Let $A = [a_{ij}]_{3\times 3}$ be a matrix such that $AA^T = 4I$ and $a_{ij} + 2c_{ij} = 0$ where c_{ij} is the cofactor of $a_{ij}\ \forall$ i & j, I is the unit matrix of order 3 and A^T is the transpose of the matrix A.
If $\begin{vmatrix} a_{11}+4 & a_{12} & a_{13} \\ a_{21} & a_{22}+4 & a_{23} \\ a_{31} & a_{32} & a_{33}+4 \end{vmatrix} + 5\lambda \begin{vmatrix} a_{11}+1 & a_{12} & a_{13} \\ a_{21} & a_{22}+1 & a_{23} \\ a_{31} & a_{32} & a_{33}+1 \end{vmatrix} = 0$, then $\lambda = \frac{a}{b}$ where a and b are coprime positive integers then the value of a + b is _____

60. If $x^3 - ax^2 + bx + c = 0$ has the roots $\alpha^2+\beta^3+\gamma^4$, $\beta^2+\gamma^3+\alpha^4$ and $\gamma^2+\alpha^3+\beta^4$ where α, β, γ are the roots of $x^3 - x^2 - 1 = 0$ then the value of a + b + c is equal to _____

Space for rough work